Question ID: 2eef7e61

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Easy

Question

The graph of the function f is a line in the xy-plane. If the line has slope $\frac{3}{4}$ and $f(0)=3$, which of the following defines f?

Answer

A. $f(x)=\frac{3}{4}x-3$

B. $f(x)=\frac{3}{4}x+3$

C. $f(x)=4x-3$

D. $f(x)=4x+3$

Correct Answer: B

Rationale

Choice B is correct. The equation for the function f in the xy-plane can be represented by $f(x)=mx+b$, where m is the slope and b is the y-coordinate of the y-intercept. Since it's given that the line has a slope of $\frac{3}{4}$, it follows that $m=\frac{3}{4}$ in $f(x)=mx+b$, which yields $y=\frac{3}{4}x+b$. It's given that $f(0)=3$. This implies that the graph of the function f in the xy-plane passes through the point $(0,3)$. Thus, the y-coordinate of the y-intercept of the graph is 3, so $b=3$ in $f(x)=\frac{3}{4}x+b$, which yields $f(x)=\frac{3}{4}x+3$. Therefore, the equation $f(x)=\frac{3}{4}x+3$ defines the function f.

Choice A is incorrect and may result from a sign error for the y-intercept. Choice C is incorrect and may result from using the denominator of the given slope as m in $f(x)=mx+b$, in addition to a sign error for the y-intercept. Choice D is incorrect and may result from using the denominator of the given slope as m in $f(x)=mx+b$.